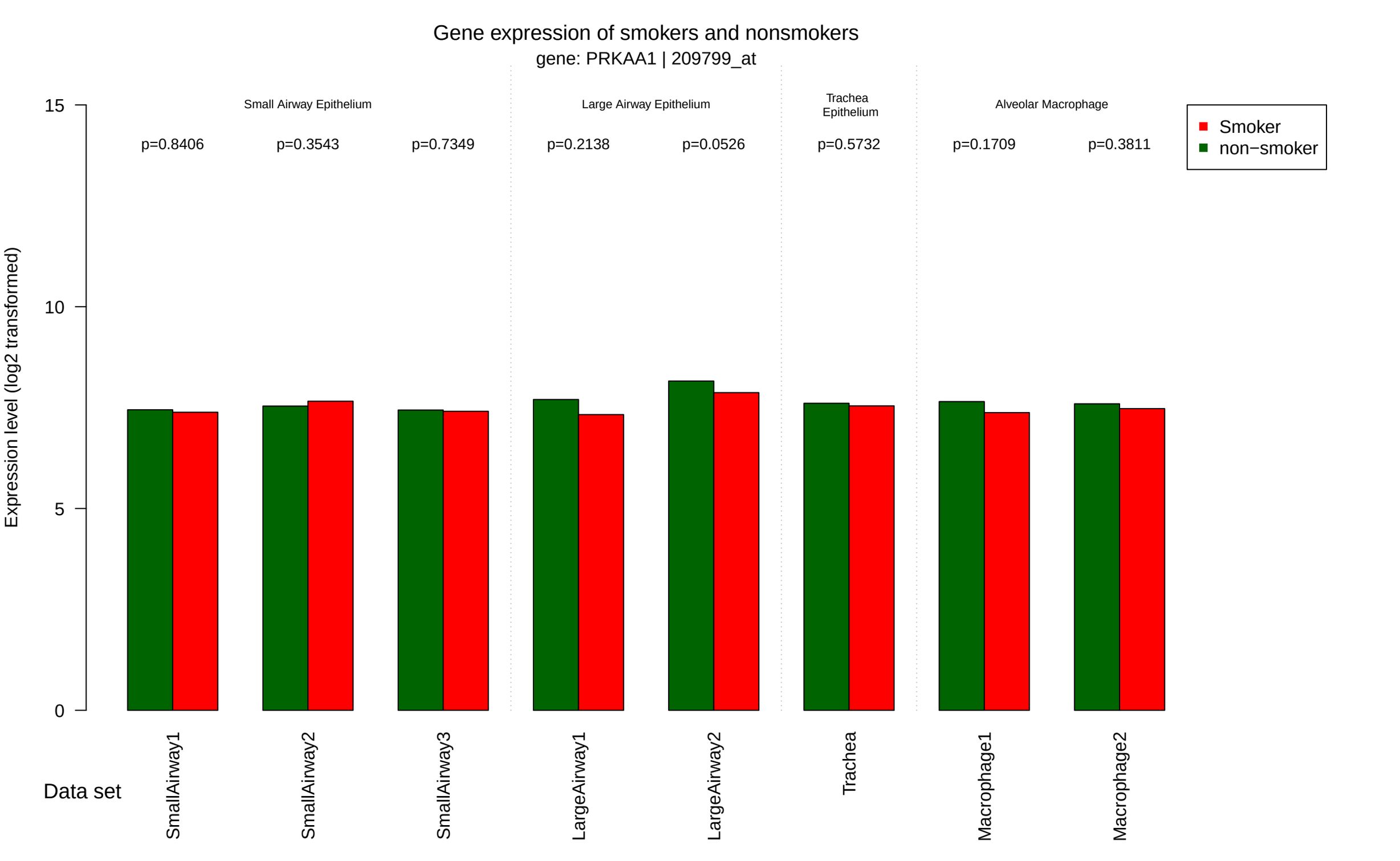
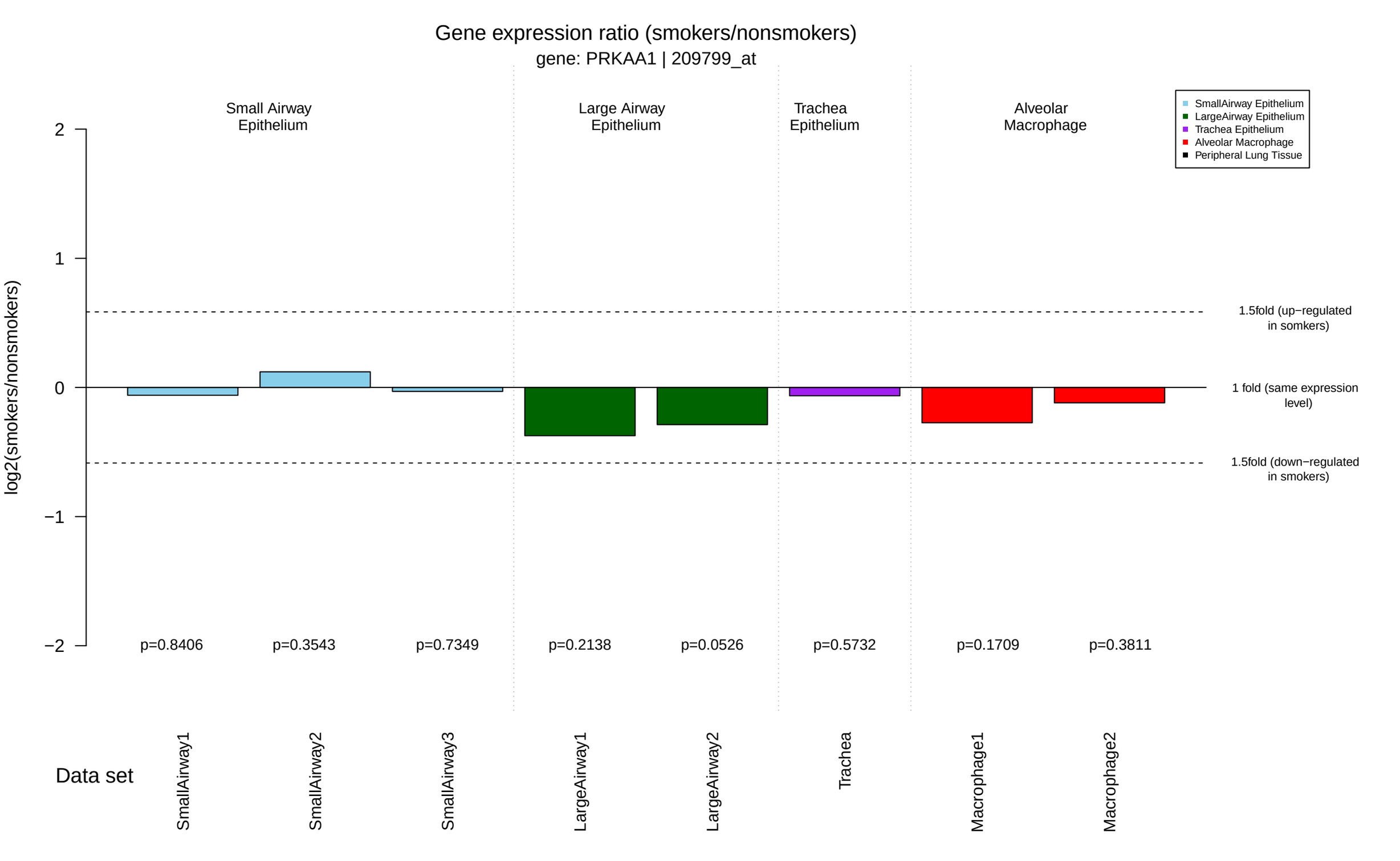
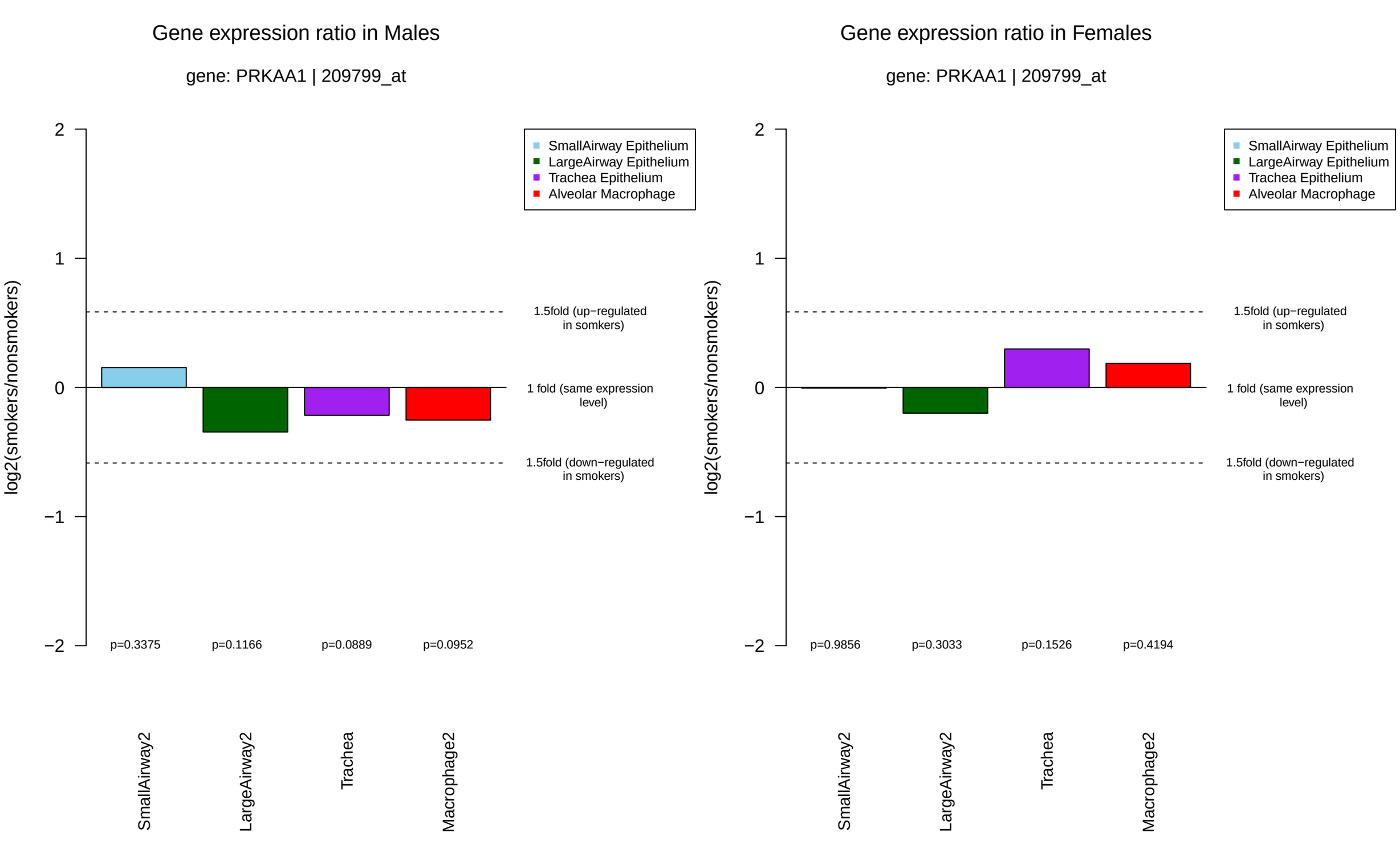
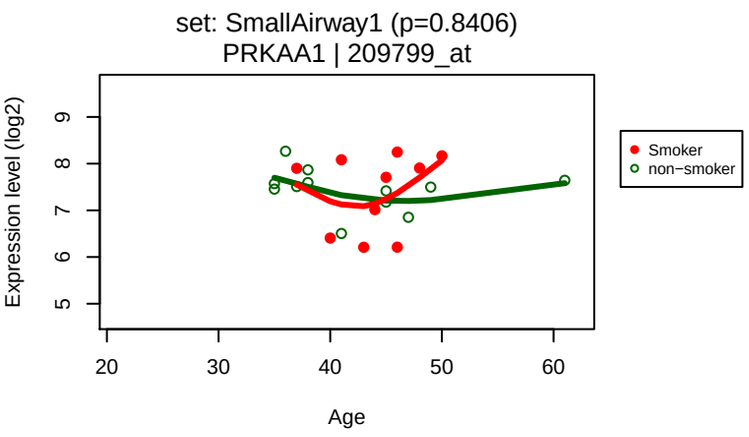
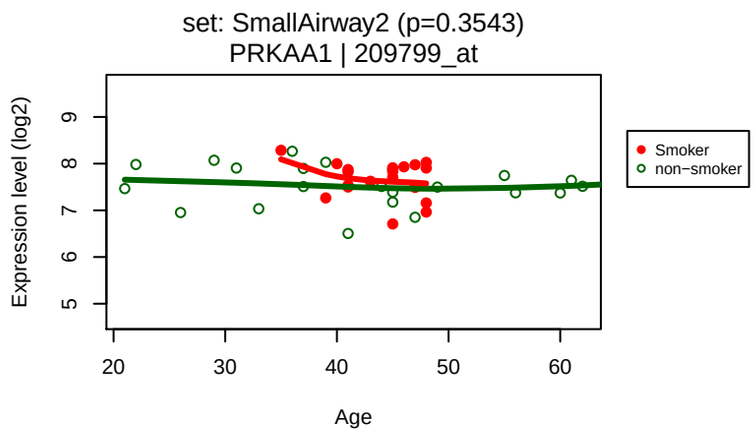
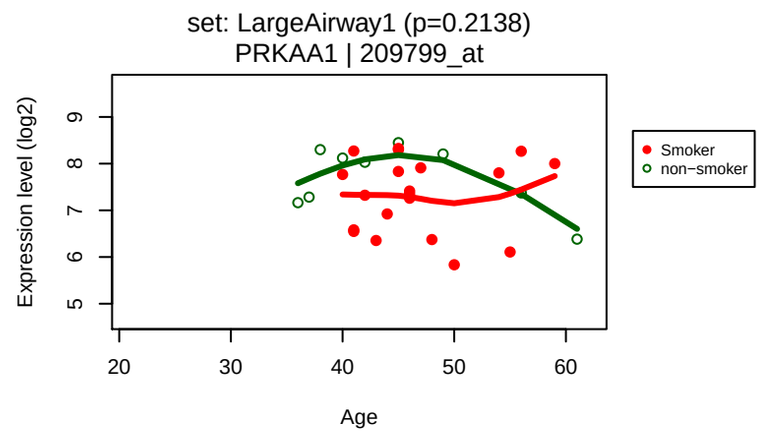

Gene expression of smokers and nonsmokers
gene: PRKAA1 | 209799_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8406
p=0.3543
p=0.7349
p=0.2138
p=0.0526
p=0.5732
p=0.1709
p=0.3811
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

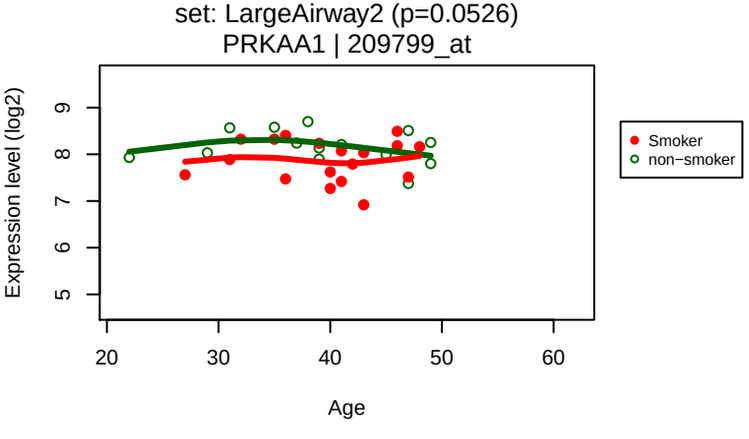

Gene expression ratio (smokers/nonsmokers)
gene: PRKAA1 | 209799_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8406
p=0.3543
p=0.7349
p=0.2138
p=0.0526
p=0.5732
p=0.1709
p=0.3811
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

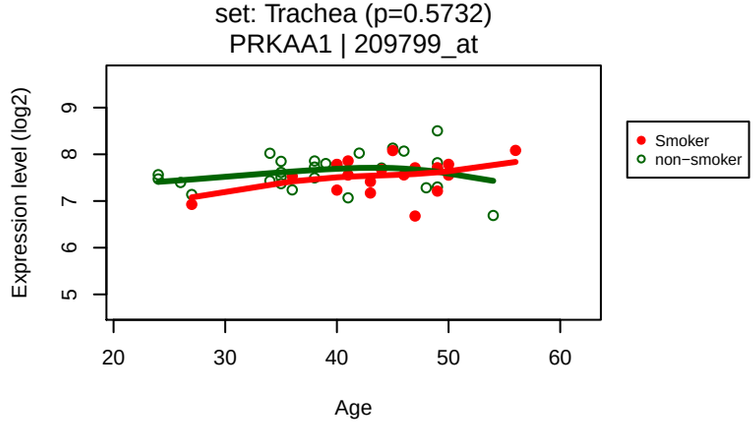

Gene expression ratio in Males
gene: PRKAA1 | 209799_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3375
p=0.1166
p=0.0889
p=0.0952
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PRKAA1 | 209799_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9856
p=0.3033
p=0.1526
p=0.4194
SmallAirway2
LargeAirway2
Trachea
Macrophage2

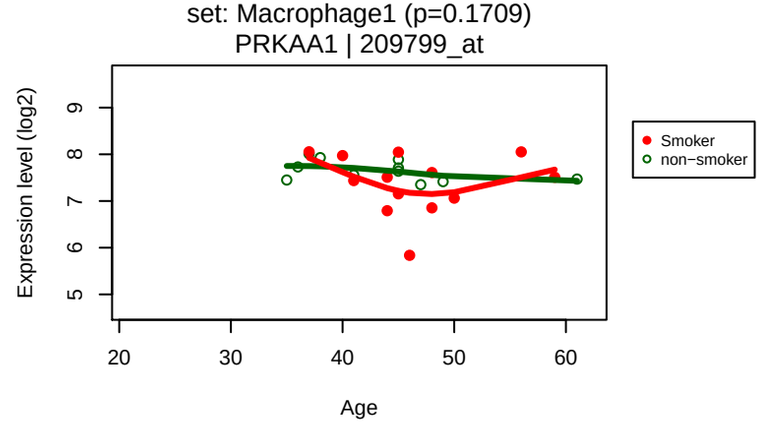

set: SmallAirway1 (p=0.8406)
PRKAA1 | 209799_at
Expression level (log2)
Age
Smoker
non−smoker

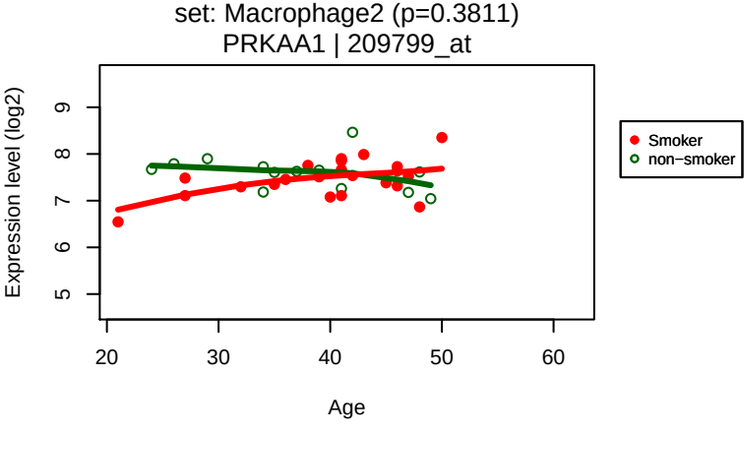

set: SmallAirway2 (p=0.3543)
PRKAA1 | 209799_at
Expression level (log2)
Age
Smoker
non−smoker

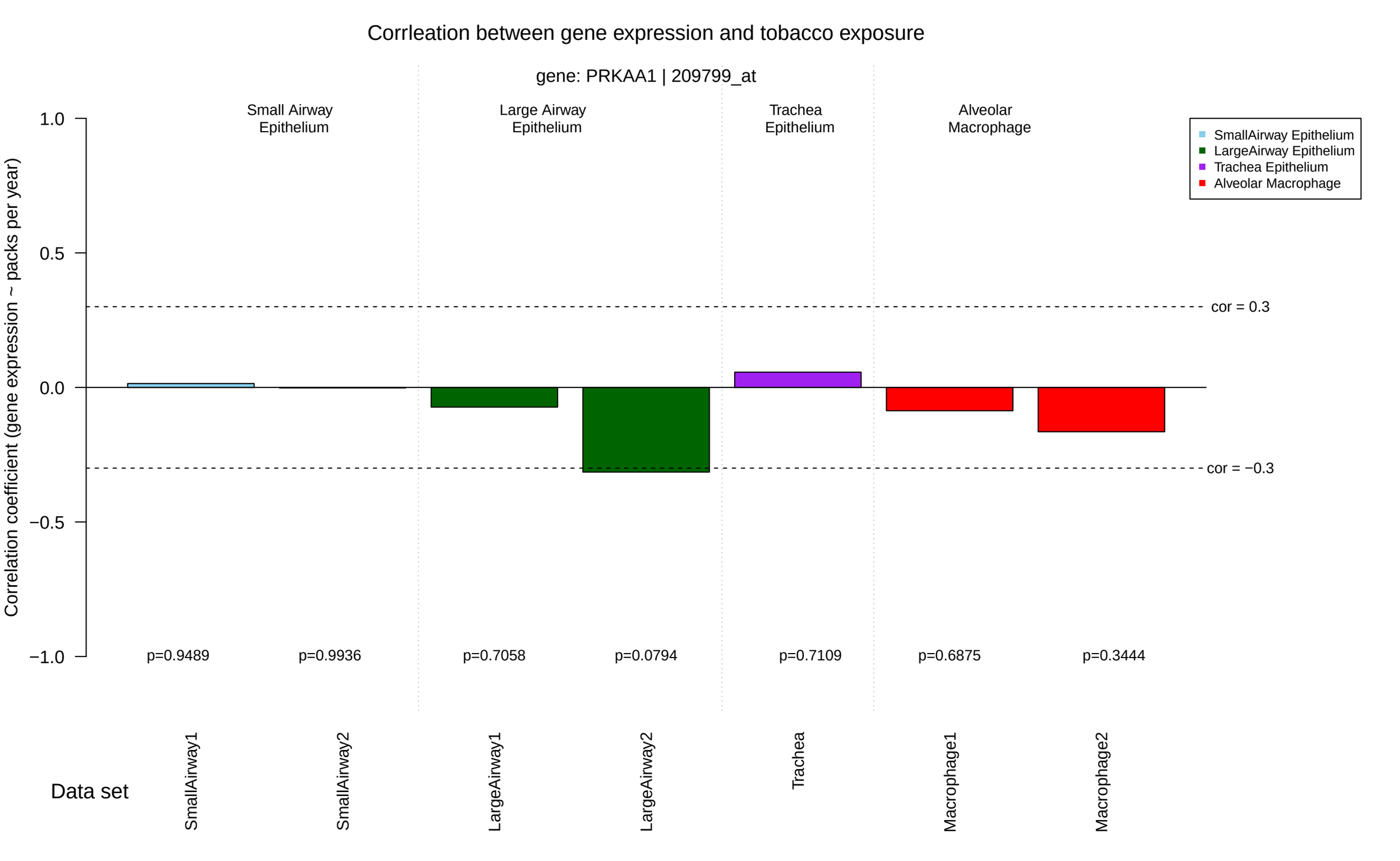

set: LargeAirway1 (p=0.2138)
PRKAA1 | 209799_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0526)
PRKAA1 | 209799_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5732)
PRKAA1 | 209799_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1709)
PRKAA1 | 209799_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3811)
PRKAA1 | 209799_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PRKAA1 | 209799_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9489
p=0.9936
p=0.7058
p=0.0794
p=0.7109
p=0.6875
p=0.3444
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2